Introduction :-

A database Management System (DBMS) is a software system that is designed to manage and organize data in a structured manner. It allows users to create, modify, and query a database, as well as manage the security and access controls for that database.

DBMS provides an environment to store and retrieve the data in convenient and efficient manner.

The software which is used to manage databases is called Database management system (DBMS). for example, MySQL, Oracle etc, are the popular commercial DBMS used in different applications. DBMS allows users the following tasks

- **Data Definition:** It helps in the creation, modification and removal of definitions that define the organization of data in the database.
- **Data updation:** It helps in the insertion, modification and deletion of the actual data in the database.
- **Data Retrieval:** It helps in the retrieval of data from the data base which can be used by applications for various purposes.
- **user Adminstration:** It helps in registering and mointoring users, enforcing data security, mointoring performance; maintaining data integrity, dealing with concurrency control, and recovering information corrupted by unexpected failure.

Key features of DBMS :

- **Data Modeling :-** A DBMS provides tools for creating and modifying data models, which define the structure and relation ships of the data in a database.
- **Data storage and retrieval :-** A DBMS is responsible for storing and retrieving data from the database, and can provide various methods for searching and querying the data.

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- Concurrency control: A DBMS provides mechanisms for controlling concurrent access to the database, to ensure that multiple users can access the data without conflicting with each other.
- Data integrity and security: A DBMS provides tools for enforcing data integrity and security constraints, such as constraints on the values of data and access controls that restrict who can access the data.
- Backup and recovery: A DBMS provides mechanisms for backing up and recovering the data in the event of a system failure.
- DBMS can be classified into 2 types:
- Relational Database Management System (RDMS):- Data is organized in the form of tables of each table has a set of rows and columns. The data are related to each other through primary and foreign keys.
- Non-Relational Database Management system (NoSQL or Non-sql):- Data is organized in the form of key value pairs, documents, graphs, or column-based. these are designed to handle large-scale, high performance scenarios.

Applications of DBMS:-

- Enterprise information: sales, accounting, human resources, manufacturing, online retailers.
- Banking and Finance sector:- Banks maintaining the customer, details, accounts loans, banking transactions, credit card transactions. Finance: storing the information about sales and holdings. purchasing of financial stocks and bonds.
- university: Maintaining the information about student course enrolled.
- Airlines: Reservations and schedules
- Telecommunications: prepaid, postpaid bill maintainance.

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Advantages of DBMS:-

- **Data Independence:** Application programs should not, ideally exposed to details of data representation and storage. The DBMS provides an abstract view of the data that hides such details.
- **Efficient data Access:** A DBMS utilizes a variety of sophisticated technology to store & retrieve data efficiently. this feature is especially important if the data is stored on external storage devices.
- **Data Integrity security:** If data is always accessed through the DBMS, the DBMS can enforce integrity constraints.
eg: Before inserting salary information for an employee, the DBMS can check that the department budget is not exceeded Also, it can enforce access controls that govern what data is visible to different classes of users.
- **Data Admistration:** when several users share the data, centralizing the adminstration of data can offer significant improvements.
- **Concurrent Access & crash recovery:** A DBMS concurrent access to the data in such a manner that users can think of the data as being accessed by only one user at a time. further, the DBMS protects users from the effects of system failures.
- **Reduced Application Development Time:** The DBMS supports important functions that are common to many application accessing data in the DBMS. This in conjunction with the high-level interface to the data, facilitates quick application development.

- DBMS applications are robust and many important tasks are handled by the DBMS.

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